Lilac Molecular Laboratory
Cytogenetics laboratory

Chromosome Analysis, Blood

Patient Name: Elias Oyelaran
Referring Physician: Leah Brennan, MS, CGC
Specimen #: 40363228 **Client #:** 91408
Patient ID: 00566688-5
DOB: 30.3.1997 Date Collected: 18.10.2024
Sex: Male Date Received: 25.10.2024
SSN: ***-**-**** Lab ID: 353535
Specimen Type: **Whole blood (sodium heparin)**
Indication: Developmental delay

Maple Grove Family Medicine
6354 Harbor Road
Lakewood, NJ 00126

Metaphases Counted:	20	**Number of Cultures:**	2	**Banding Technique:**	GTW
Metaphases Analyzed:	5			**Banding Resolution:**	450
Metaphases Karyotyped:	2			**Dept. Section:**	C8

RESULTS: 46,XY
Normal male karyotype

INTERPRETATION:

The karyotype is normal and shows a normal male pattern of chromosomes. No large chromosome change was found.

A karyotype can detect large changes in chromosome number and structure. It cannot detect small changes in single genes, or small deletions and duplications. The normal result therefore rules out only the large chromosome changes that a karyotype can see.

The result does not explain the reason for testing. A chromosomal microarray or exome or genome sequencing may be needed if the developmental delay still needs an explanation.

RECOMMENDATION:

Use the karyotype result together with the clinical findings to decide whether further testing is needed.

Consider a chromosomal microarray or exome or genome sequencing if the developmental delay is still unexplained, because these tests can detect smaller changes.

Refer the patient to a clinical geneticist to review the result with the reported features.

We recommend genetic counseling for the family.

COMMENT:

Routine chromosome analysis, as used here, can miss subtle rearrangements and low-level mosaicism, and it cannot see the small deletions and duplications that a microarray may detect.

Signed: Paul Nakashima, PhD, FACMG Date: 7.12.2024

ABOUT THIS RESULT

The test looks for large chromosome changes that are linked to developmental delay, intellectual disability, growth differences and birth features. Such changes include extra or missing chromosomes and large rearrangements of chromosome parts.

Developmental delay affects a meaningful share of children. Large chromosome changes account for only a small part of these cases, and most cases have no chromosome cause that a karyotype can find.

Developmental delay has many causes, including chromosome changes, changes in single genes and environmental factors. A karyotype looks only for large chromosome changes.

INFORMATION FOR THE PATIENT

The chromosome test looked at the full set of chromosomes in the blood cells. The chromosomes look normal in number and in shape.

This test can find large changes in chromosomes, but it cannot find small changes in single genes. A normal result does not rule out a genetic cause for the developmental delay.

Your doctor can talk with you about whether other genetic tests are a good next step.